

Question number	Answer	Additional guidance	Mark
6(a)(i)	The correct answer is B <i>A is incorrect because the head is a phosphate and the tails are fatty acids</i> <i>C is incorrect because the head is a phosphate and the tails are fatty acids</i> <i>D is incorrect because the heads are hydrophilic and the tails are hydrophobic</i>		(1)

Question number	Answer	Additional guidance	Mark
6(a)(ii)	An answer that includes four of the following points: Similarities: • both have proteins (1) • both have {phospholipid / phospholipid bilayer} (1) Differences: • FMM has proteins {embedded / intrinsic}, but DDM has proteins {outside / in a layer/ extrinsic} (1) • FMM has cholesterol but DDM does not (1) • FMM has {glycoproteins / glycolipids} but DDM does not (1)	DO NOT PIECE TOGETHER FROM TWO DESCRIPTIONS ACCEPT description of embedded proteins e.g. channel proteins DDM proteins are not embedded	(4)

Question number	Answer	Additional guidance	Mark
6(b)(i)	An answer that includes two of the following points: • there could be (uncontrolled) movement of {substances / named substances} (into / out of cell) (1) • cell cannot maintain concentration gradients of substances (1) • credit consequence of this (1) • loss of cell {recognition / adhesion} / loss of membrane-embedded molecules (1)	ACCEPT cytoplasm could leak out molecules will not be able to pass through {active transport / facilitated diffusion} could not take place bacteria / organelles ACCEPT {solute / water / osmotic} potential e.g. cannot maintain ATP levels, cell could {shrivel up / burst}, no oxygen for (aerobic) respiration	(2)

Question number	Answer	Additional guidance	Mark
6(b)(ii)	An explanation that includes the following points: • because the phospholipids (and proteins) {can move / are fluid} (within the membrane) (1) • therefore {phospholipids / fatty acids} can interact (hydrophobically) (1)	ACCEPT bond Van der Waals forces / London forces / hydrophobic interactions vesicles fusing with membrane (by exocytosis / endocytosis)	(2)

Question number	Answer	Additional guidance	Mark
6(b)(iii)	An answer that includes two of the following points: • more {membrane / phospholipid / protein / glycoprotein / glycolipid / cholesterol} needs to be made (1) • membrane needs restructuring / proteins added to membrane (1) • increase rate of {respiration / ATP synthesis / release of energy} (1)	ACCEPT {channel / carrier} proteins ACCEPT {cholesterol / glycolipids / glycoproteins} need to move into place	(2)